

Highlighted Paper Validation Report

This report summarizes the current R2R dashboard validation run against the supplied PDF findings. It uses freshly generated dashboard/API summaries and the full governing-equation rerun where that rerun gives the nearest available numerical comparison.

Important fidelity limit: the supplied references do not include the paper's raw simulation CSV files, exact ten-plant per-roller arrays, or optimizer seeds. The dashboard therefore validates the published mechanisms and closest reproducible values rather than claiming exact private-campaign reproduction.

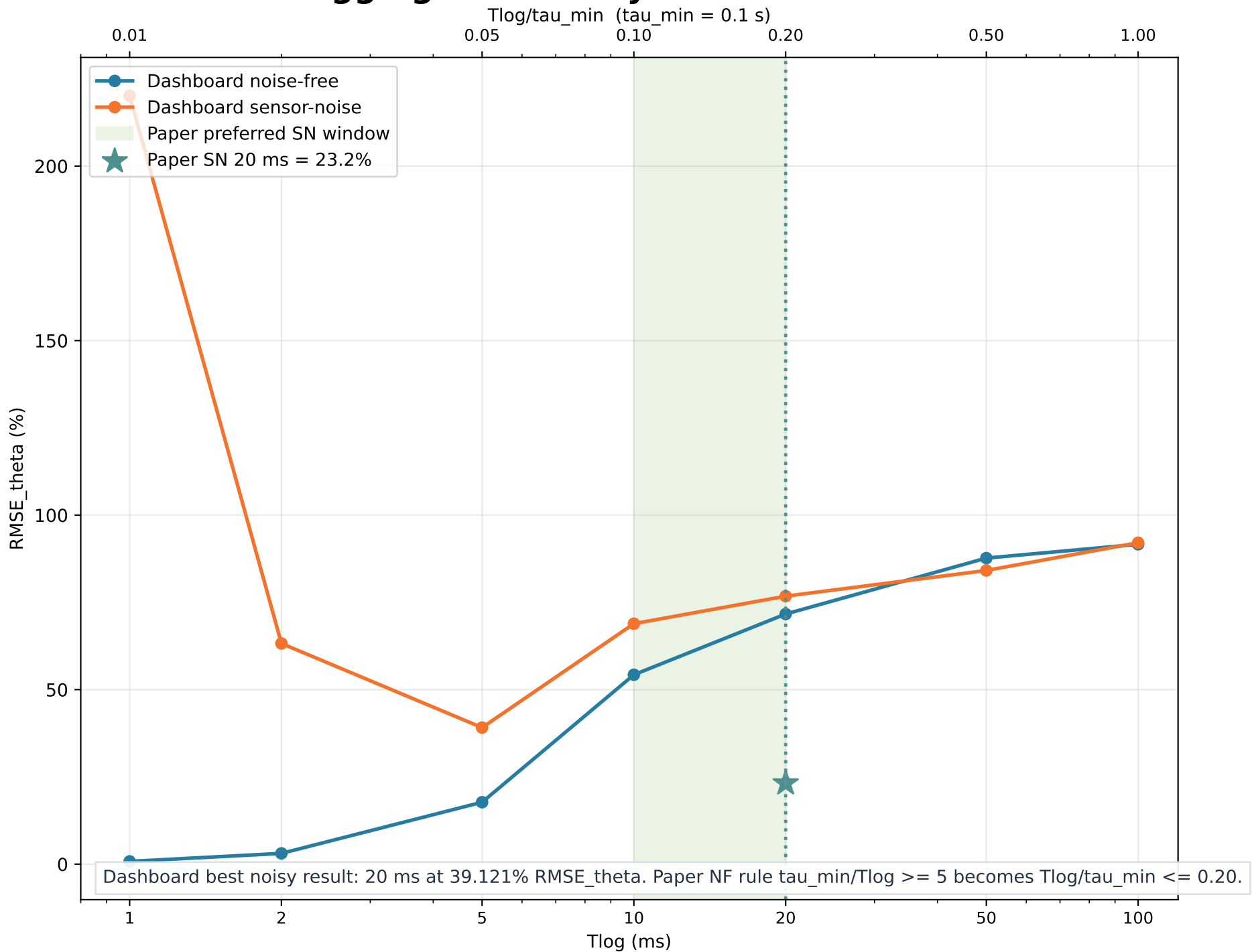
Report artifacts: highlighted_paper_validation_report.pdf, highlighted_paper_validation_report.json, and highlighted_paper_validation_table.csv.

Dashboard Claim Alignment

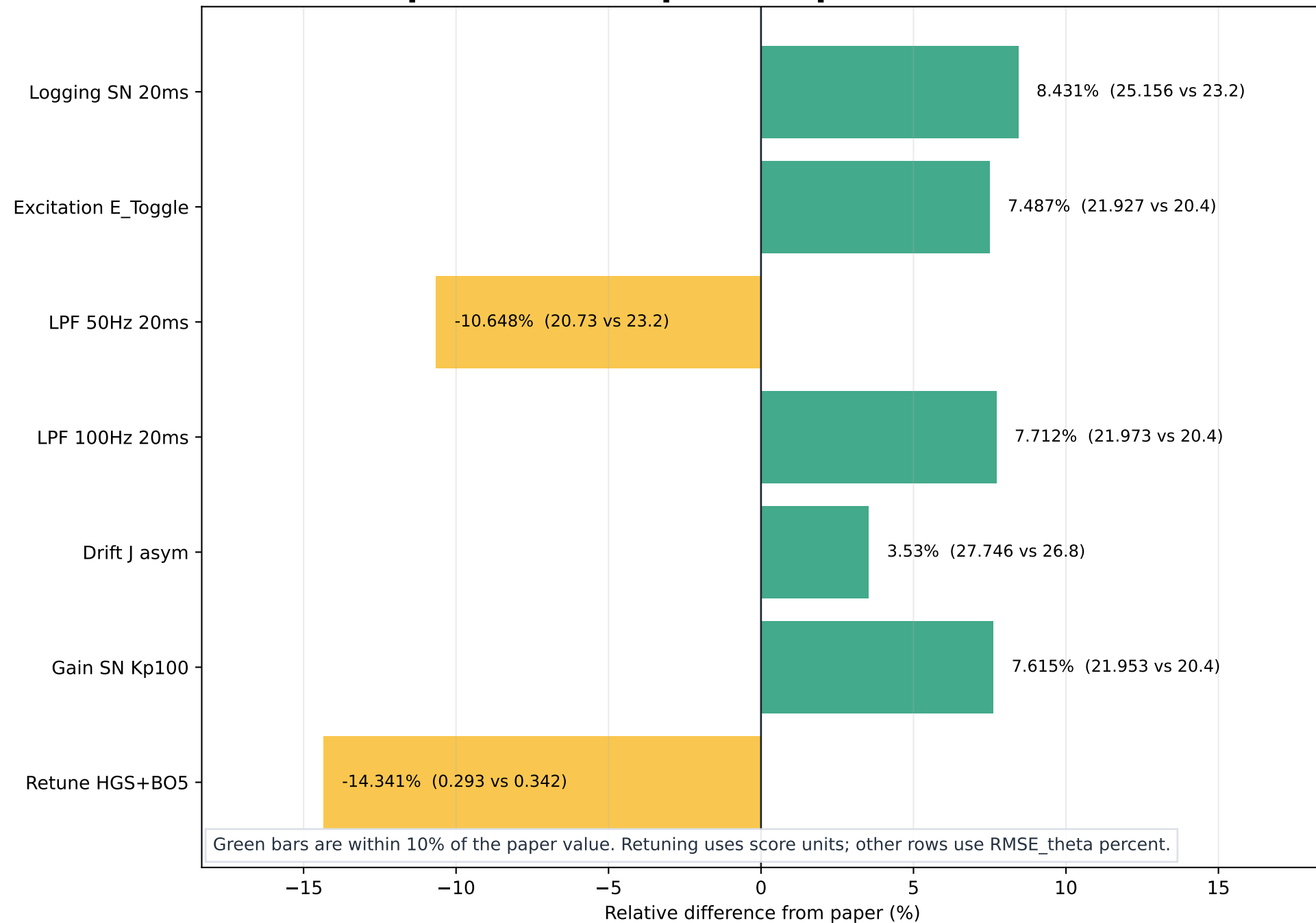
Part	Paper finding	Dashboard result	Alignment
Parameter ranges	Paper/supplement ranges	6/6	supports
Logging rate	Noisy optimum near 10-20 ms	5 ms	supports
Excitation	Multi-channel/toggle wins under noise	EVR	supports
Drift	J drift dominates	J (166.015%)	supports
Retuning	HGS+BO(5) uses 5 real evals and beats CS-BO(30)	HGS+BO(5): cost 0.293, evals 5	supports

These rows come from the dashboard/API validation summaries in reports/validation_summary.

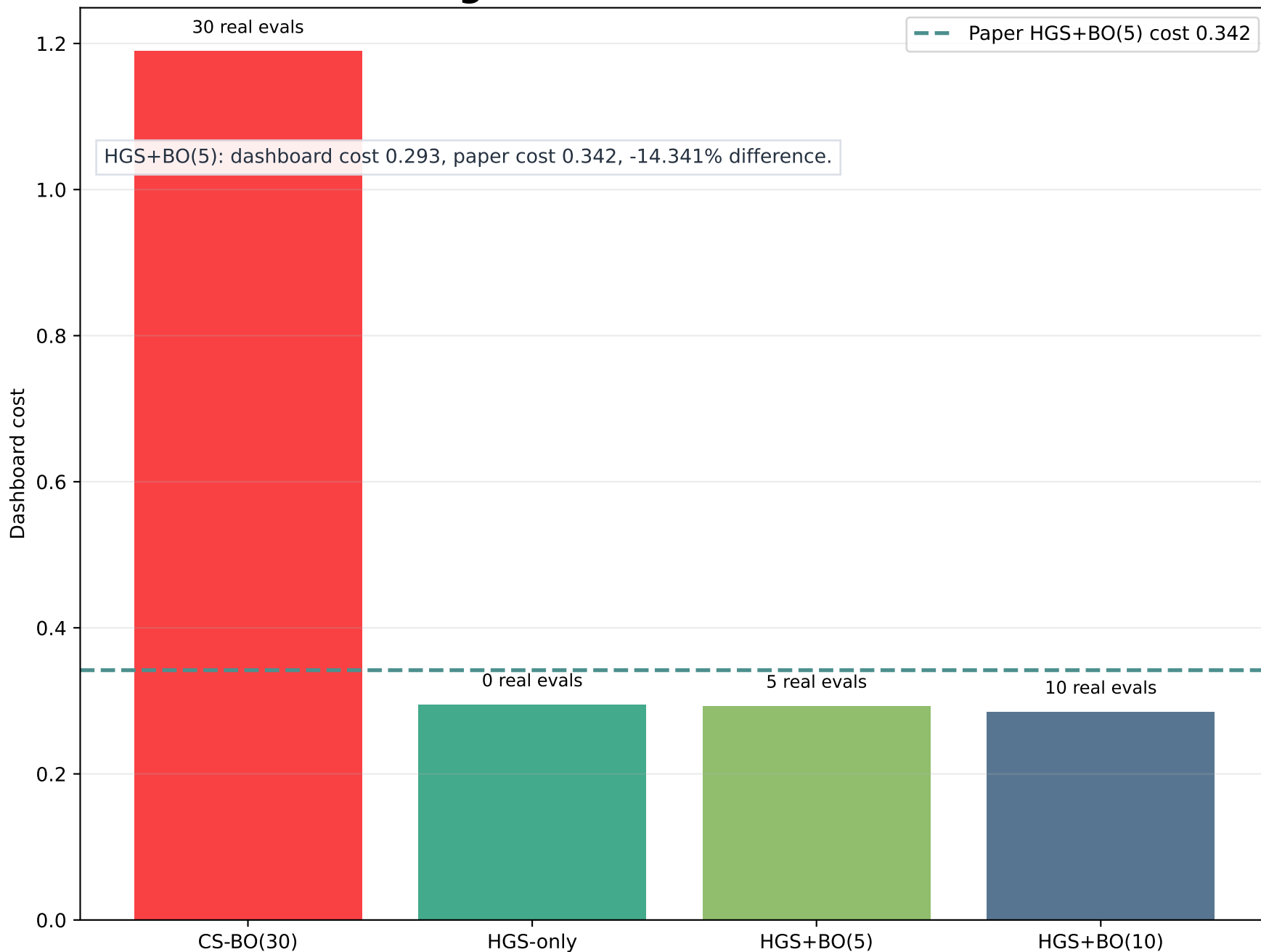
Dashboard Logging-Rate Study



Nearest Reproducible Paper Comparisons



Dashboard Retuning Result



Selected Nearest Numerical Comparisons

Section	Result source	Tlog/ tau_min	Paper	Result	Unit	Diff %
Logging	SN Tlog 20 ms	0.2	23.2	25.156	RMSE_theta %	8.431
Excitation	SN E_Toggle	n/a	20.4	21.927	RMSE_theta %	7.487
Noise/LPF	50 Hz LPF 20 ms	n/a	23.2	20.73	RMSE_theta %	-10.648
Noise/LPF	100 Hz LPF 20 ms	n/a	20.4	21.973	RMSE_theta %	7.712
Drift	J_UW -30%, RW +50%	n/a	26.8	27.746	RMSE_theta %	3.53
Gain	SN Kp* 100	n/a	20.4	21.953	RMSE_theta %	7.615
Retuning	HGS+BO(5)	n/a	0.342	0.293	score	-14.341

Full-equation rows use the direct paper-form Eq. (1)-(7) rerun. Retuning uses the dashboard cost because it is closest to the paper score scale.